

PNEUMONIA DIAGNOSIS DETECTION AND LOCALIZATION

Weakly Supervised Grad-CAM on Chest X-rays

Abstract. This mini-project develops a weakly supervised framework that classifies chest X-rays as pneumonia/no pneumonia and provides an approximate localization of the image regions influencing the prediction. A DenseNet121 model initialized with ImageNet weights is trained only with image-level labels from the RSNA Pneumonia Detection Challenge. Grad-CAM is then converted into candidate bounding boxes using adaptive thresholding, morphological cleanup, and connected components. The final 224x224 run improved discrimination and substantially improved localization relative to the 128x128 baseline, while remaining a research prototype rather than a clinically validated diagnostic system.

1. Problem and Motivation

Chest X-ray classification can indicate whether pneumonia is present, but classification alone does not show where the model is focusing. Pixel- or box-level annotations are expensive, so the project targets weakly supervised localization: use image-level pneumonia labels for learning, then derive an approximate region of interest with Grad-CAM. Ground-truth RSNA boxes are reserved for evaluation and are not training targets.

2. Dataset and Split

The experiment uses the labeled RSNA training partition. There are 30,227 annotation rows corresponding to 26,684 unique image IDs after grouping annotation rows by image. The set contains 6,012 pneumonia-positive and 20,672 negative images. A reproducible 80/10/10 image-level split with seed 42 gives 21,347 train, 2,668 validation, and 2,669 test images.

Images	Positive	Negative	Split
26,684	6,012	20,672	80/10/10

3. Approach and Implementation

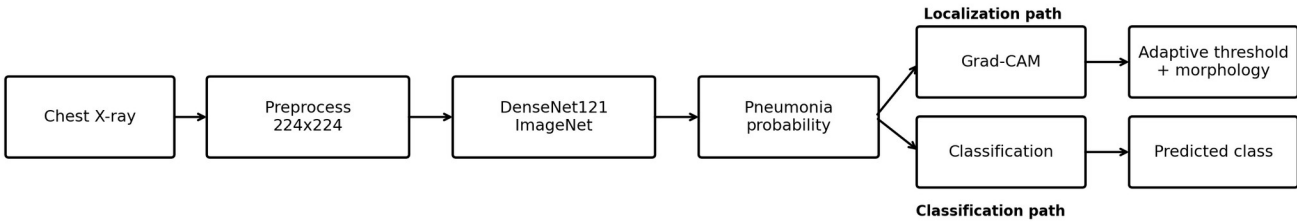


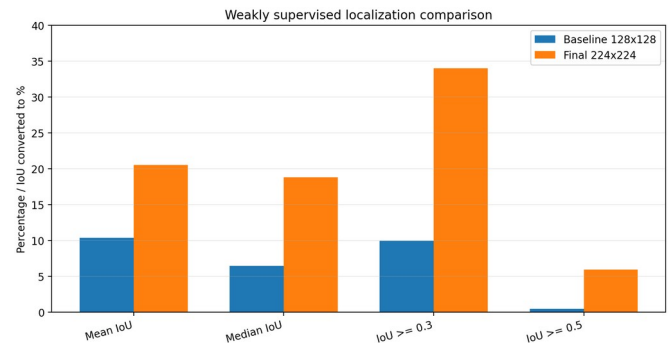
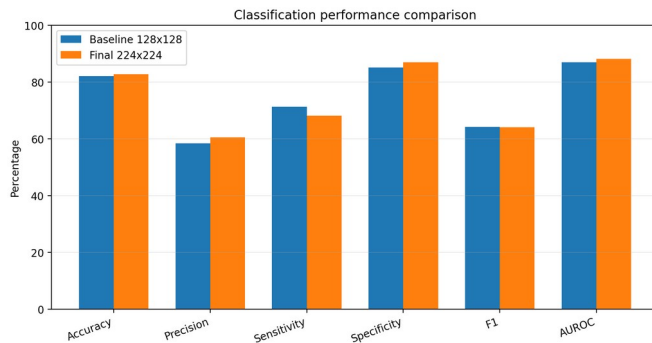
Figure 1. End-to-end weakly supervised detection and localization pipeline.

Component	Final configuration
Classifier	DenseNet121, ImageNet pretrained
Input	224x224
Loss	Weighted binary cross-entropy
Optimizer	AdamW, learning rate 1e-4
Training	Batch size 8; up to 10 epochs; early stopping patience 3; best epoch 3; 6 epochs completed
Localization	Grad-CAM -> adaptive threshold -> morphology -> 8-connected components -> candidate boxes
Thresholding	Selected using validation F1 and then fixed for test; final operating threshold 0.70
Evaluation	Classification metrics + AUROC; mean/median IoU and IoU >= 0.3/0.5

4. Results

The final 224x224 run is compared with the preserved 128x128 baseline. The same image-level weak supervision is used in both runs; the final experiment tests whether higher spatial resolution and longer training improve performance.

Metric	Baseline 128x128	Final 224x224
Accuracy	82.02%	82.73%
Precision	58.34%	60.44%
Sensitivity / Recall	71.31%	68.16%
Specificity	85.14%	86.98%
F1	64.18%	64.07%
AUROC	86.97%	88.15%
Mean IoU	0.1038	0.2051
Median IoU	0.0645	0.1880
IoU ≥ 0.3	9.95%	34.00%
IoU ≥ 0.5	0.50%	5.97%



5. Discussion and Limitations

- Classification is reasonably useful for a student research prototype: AUROC increases from 0.8697 to 0.8815 in the final run, while specificity also improves. Sensitivity and F1 are slightly lower, so the higher-resolution run does not dominate every classification metric.
- Localization improves strongly with 224x224 input: mean IoU rises from 0.1038 to 0.2051 and the fraction reaching IoU ≥ 0.3 rises from 9.95% to 34.00%. However, only 5.97% reach IoU ≥ 0.5 , so the heatmaps remain coarse.
- The method is weakly supervised: Grad-CAM identifies influential regions rather than producing validated lesion segmentation. Evaluation is limited to the RSNA labeled training partition, so external generalization is not established.
- The primary localization score assigns IoU = 0 to classifier false negatives; an independent localization score is also reported to separate classification errors from localization quality.

6. Conclusion

The project demonstrates an end-to-end weakly supervised pneumonia detection and localization pipeline using DenseNet121 and Grad-CAM. The 224x224 final experiment improves AUROC and substantially improves localization overlap relative to the 128x128 baseline. The result supports the value of preserving more spatial information, but the localization remains an approximate explanatory output and requires stronger validation before any clinical interpretation.

References

- [1] Shih-Cheng Huang et al., "Weakly Supervised Pneumonia Localization," reference paper supplied for the mini-project.
- [2] RSNA, "Pneumonia Detection Challenge," Kaggle dataset and annotation files.
- [3] Project repository: Pneumonia Diagnosis Detection and Weakly Supervised Localization using chest X-rays.

Research prototype - not a clinical diagnostic device.